A Bounded Failure Window in a Miscalibrated Precision-Weighted Relay

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Abstract

We study a linear constant-gain filter, called here a *relay*, whose gain is set by the one-step Bayesian weight $\Pi/(\Pi + \Pi_0)$ rather than by the steady-state Kalman–Bucy gain. We ask a narrow question: at which observation precisions Π does its stationary error variance exceed a fixed tolerance R^2?

The variance exceeds its zero-precision value F exactly when the gain exceeds $2F\Pi$, whereas the Kalman–Bucy gain never exceeds $F\Pi$. As a consequence, for gains with natural limits, some fixed tolerance fails only on a **bounded set** of precisions away from zero if and only if $\sup_\Pi G/\Pi > 2F$. The variance then stays below the tolerance at low precision, exceeds it at intermediate precision, and falls below it again at high precision, where the correction dominates the intrinsic dynamics and the error levels off at a positive limit H; the optimal variance instead falls to zero. The steady-state Kalman–Bucy filter, whose error variance decreases monotonically, cannot produce such a set.

For the Bayesian-weight gain, a window exists if and only if the over-weighting ratio κ, the gain's initial slope divided by the optimal slope, exceeds 2; the peak is then unique and analytically located. The effect is small unless the miscalibration is large: $\kappa \leq 2$ gives no peak, $\kappa = 3$ raises the variance by at most 3.6%, and in the worked example failing a tolerance of five times the floor requires $\kappa > 50$. A prior matched to the coordinate's own precision gives $\kappa = c$, so a weakly coupled relay ($c < 2$) with a correctly set prior cannot open a window.

The excess identity extends exactly to coloured sensory noise and to discrete time, where it becomes $L/(2 - L) > F\Pi_d$ for gain L and per-sample precision Π_d. Symbolic and simulation checks, specified in advance, confirm the closed forms. Exploratory checks show that the window persists under signal drift, and that sigmoidal gains can open one with no initial over-weighting.

The contribution is a characterization of the failure set within this model class, not a new mechanism: that an over-gained filter can do worse than no filtering is familiar.

1. Introduction

Mismatched filters are an old subject. Heffes [7] and Friedland [3] analysed the excess error of a Kalman filter run with an incorrect noise model or gain. Nishimura [10] and Griffin and Sage [6] derived the corresponding actual-error equations and bounds for continuous-time filters.

Later work covers related ground:

- the actual error covariance of discrete-time filters under incorrect noise covariances [12];

- recursions for the actual error variance when the stochastic model is misspecified [14];
- rankings of ideal, filter-calculated and true mean-squared error under mismatched noise covariances [5];
- explicit mean-squared error for mismatched linear filters, used to select process noise [2].

That literature answers the question *how much worse is a mismatched filter at a given operating point?*

This note asks a different question: *as observation precision is swept, where does a given relay fail a fixed tolerance?* One might expect a threshold: failure that begins at some precision and then persists, or that vanishes as precision grows. The result here is that, within linear scalar filters of this kind, a third shape is also possible:

> ***A fixed tolerance can fail on a bounded set of precisions away from zero only if the gain somewhere exceeds*** $2F\Pi$***, a level the steady-state Kalman–Bucy gain never reaches.***

The mechanism is not new. A filter whose gain is too high admits more measurement noise than the damping it buys is worth, and can do worse than no filtering at all; this is the familiar noise-versus-responsiveness trade-off of constant-gain filters. What is new is a characterization: the exact shape of the failure set as precision varies, a necessary-and-sufficient condition for a bounded window, and a proof that the optimal filter cannot produce one.

A stylized setting. Consider an onboard estimate of a quantity that fluctuates about a nominal value, corrected by an external fix whose quality varies with conditions (satellite geometry, signal strength). Suppose the correction weight is set by the one-step Bayesian rule $\Pi/(\Pi+\Pi_0)$ with a fixed prior precision Π_0, rather than by a steady-state Kalman gain, and that the estimate must stay within an accuracy specification R^2. If Π_0 is set too low, there can be a band of intermediate fix quality in which the corrected estimate violates the specification, even though both poorer and better fixes are acceptable, and even though the estimate would meet the specification with no correction at all. This is a qualitative possibility with a high threshold, not an expected operating regime: Section 4 shows that the band matters only when Π_0 is set far too low.

The question first arose in models of internally generated neural activity coupled weakly to sensory input, which is the source of the term *relay*. Nothing below depends on that setting.

2. Model

A scalar coordinate $z(t)$ relaxes toward a target z^* with rate λ_{sim} and is coupled to a real-time signal $s(t) = z^* + \xi_s(t)$:

$$\dot{z} = -\lambda_{\text{sim}}(z - z^*) + \varepsilon Q K_s \left(s(t) - z\right) + \xi_{\text{sim}}(t), \qquad K_s = \frac{\Pi_s}{\Pi_s + \Pi_{\text{prior}}}. \tag{1}$$

The noises are independent Gaussian white noise, with $\langle \xi_s(t)\xi_s(t')\rangle = \Pi_s^{-1}\delta(t-t')$ and $\langle \xi_{\text{sim}}(t)\xi_{\text{sim}}(t')\rangle = 2\Sigma_{\text{sim}}\delta(t-t')$.

Notation. Write $\lambda = \lambda_{\text{sim}}$, $\Sigma = \Sigma_{\text{sim}}$, $c = \varepsilon Q$, $\Pi_0 = \Pi_{\text{prior}}$ and $\Pi = \Pi_s$, the observation (sensory) precision. Let $G(\Pi)$ be the effective gain and $G_\infty = \lim_{\Pi\to\infty} G(\Pi)$. With $x = z - z^*$,

$$\dot{x} = -(\lambda + G)\, x + G\, \xi_s + \xi_{\text{sim}}.$$

Equivalence to a constant-gain filter. Let a hidden Ornstein–Uhlenbeck state h satisfy $\dot{h} = -\lambda h + \sqrt{2\Sigma}\,\dot{W}$. Observe it as $y = h + \xi_s$ at precision Π, and estimate it with the constant-gain filter $\dot{\hat{h}} = -\lambda\hat{h} + G(y - \hat{h})$. The estimation error $\hat{h} - h$ obeys the same stochastic differential equation as x, up to the sign of the process-noise term, and so has the same stationary law. Every result below is therefore a statement about the actual error variance of a constant-gain filter for this plant.

The Bayesian-weight gain. The gain $K_s = \Pi/(\Pi + \Pi_0)$ is the posterior weight given to a single observation of precision Π combined with a prior of precision Π_0: the one-step Kalman gain. It is *not* the steady-state Kalman–Bucy gain, which for this plant is $\sigma_\star^2\,\Pi$ (Proposition 3). We call $G = cK_s$, with $G_\infty = c$, the **Bayesian-weight gain**. Propositions 1 and 2 and the theorem of Section 3.4 hold for a general gain G.

Constant gain, varying schedule. At each precision Π the relay's gain is constant in time. Across the precision sweep, $G(\Pi)$ is a gain *schedule* that may vary. "Constant-gain" below always means constant in time at a given precision; the precision-blind case, in which G does not depend on Π at all, is treated separately in Section 3.8. Throughout, σ^2, M_L and S_L denote *actual* stationary error variances, never the variances a filter reports from its own model.

Three notions of "stability," kept separate.

- *Dynamical stability.* The relay is dynamically stable at every precision, since $\lambda + G > 0$.
- *Statistical confinement.* This is the object of study: the stationary variance σ^2.
- *Operational failure.* This means $\sigma^2 > R^2$ for a task-specified tolerance R^2, which is a modelling choice.

3. Results

3.1 Stationary variance

Proposition 1. The stationary variance of (1) is

$$\sigma^2(\Pi) = \frac{\Sigma + \frac{1}{2}G^2/\Pi}{\lambda + G}. \tag{2}$$

If $G \to 0$ and $G^2/\Pi \to 0$ as $\Pi \to 0$, then σ^2 tends to the **floor** $F = \Sigma/\lambda$. If $G \to G_\infty \in (0, \infty)$ as $\Pi \to \infty$, it tends to the **high-precision limit** $H = \Sigma/(\lambda + G_\infty) < F$.

Proof. x is an Ornstein–Uhlenbeck process with rate $a = \lambda + G$ and noise intensity $D = G^2/\Pi + 2\Sigma$, so its stationary variance is $D/(2a)$. □

3.2 The excess identity

Proposition 2. For any gain,

$$\sigma^2(\Pi) - F = \frac{G\left(\lambda G/(2\Pi) - \Sigma\right)}{\lambda(\lambda + G)}, \qquad \text{so} \qquad \sigma^2(\Pi) > F \iff G(\Pi) > 2F\,\Pi. \tag{3}$$

Proof. Subtract Σ/λ from (2) and collect terms. □

3.3 The optimal benchmark

Proposition 3 (steady-state Kalman–Bucy filter [9]). For this plant, the steady-state Riccati equation $0 = 2\Sigma - 2\lambda p - \Pi p^2$ gives

$$G^\star(\Pi) = p\,\Pi = -\lambda + \sqrt{\lambda^2 + 2\Sigma\Pi}, \qquad \sigma^2_\star(\Pi) = p = \frac{2\Sigma}{\lambda + \sqrt{\lambda^2 + 2\Sigma\Pi}}. \tag{4}$$

$G^\star$ also minimizes (2) over constant gains, so $\sigma^2(\Pi) \geq \sigma^2_\star(\Pi)$ for every gain.

This is standard. The remark used below is that $\sigma^2_\star$ decreases strictly from F, so

$$G^\star = \sigma^2_\star\,\Pi \leq F\Pi.$$

The optimal gain therefore stays at least a factor of 2 below the excess threshold $2F\Pi$ in (3).

3.4 A criterion for bounded failure

Three reference points. A relay's variance can be judged against three baselines, and the result depends on keeping them apart:

- *the optimal filter*, $\sigma^2_\star$: every relay is at or above it at every precision (Proposition 3), so suboptimality alone says nothing about failure;
- *no filtering*, F: the excess identity (3) says exactly when the relay is worse than this;
- *the task tolerance*, R^2: exceeding it is failure, and the window is defined against it.

The theorem below connects the second baseline to the third.

Define the failure set $\mathcal{F}(R^2) = \{\Pi > 0 : \sigma^2(\Pi) > R^2\}$. Call it a **window** if it is nonempty, bounded, and bounded away from 0; a window need not be a single interval.

Theorem. Let G be continuous, with $G \to 0$ and $G^2/\Pi \to 0$ as $\Pi \to 0$, and $G \to G_\infty \in (0, \infty)$ as $\Pi \to \infty$. Then some tolerance R^2 produces a window if and only if

$$\sup_{\Pi > 0} \frac{G(\Pi)}{\Pi} \;>\; 2F = \frac{2\Sigma}{\lambda}. \tag{5}$$

Every $R^2 \in (F, \sup \sigma^2)$ produces a window. If $\sigma^2 > F$ for all small Π, as for the Bayesian-weight gain, these are the only windows; if σ^2 approaches F from below, the tolerance $R^2 = F$ produces one as well. The optimal filter needs no appeal to the theorem: Proposition 3 shows directly that its variance decreases strictly, so its failure set is always $(0, \Pi_b)$ or empty. (Its gain grows without bound, so the theorem's bounded-gain hypothesis would not apply to it in any case.)

Proof. By Proposition 1, $\sigma^2 \to F$ as $\Pi \to 0$ and $\sigma^2 \to H < F$ as $\Pi \to \infty$.

- *If (5) holds.* By (3), $\sup \sigma^2 > F$. For $R^2 \in (F, \sup \sigma^2)$, the failure set is nonempty and, by continuity, excludes neighbourhoods of 0 and ∞.
- *If (5) fails.* Then $G(\Pi)/\Pi \leq 2F$ for every Π, so $G(\Pi) \leq 2F\Pi$, and $\sigma^2(\Pi) \leq F$ by (3). Any nonempty failure set therefore needs $R^2 < F$, and it then contains all small Π.

For the optimal filter, $G^\star/\Pi = \sigma^2_\star \leq F < 2F$, so (5) never holds; the form of its failure set follows from Proposition 3. □

In general a window may be a union of intervals. By (3), $\sigma^2 > F$ exactly where $G > 2F\Pi$. If that set has two separated components, every tolerance slightly above F (and below both local peaks of σ^2) fails on at least two separated intervals. Uniqueness holds for the Bayesian-weight gain (Proposition 4), and no tested gain produced more than one interval.

Possible versus relevant. The theorem establishes when a window is *possible*; the tolerance decides whether it *matters*. Windows occur only for R^2 between F and the peak variance, and Section 4 shows that the peak is small unless the over-weighting is large.

Windows need loose tolerances. A window requires $R^2 \geq F$ (strictly $R^2 > F$ for the Bayesian-weight gain), so it can arise only when the tolerance is at least as loose as the variance with no correction at all. The uncorrected coordinate then already meets the specification, and switching on the precision-weighted correction breaks it within a band of precision. Any tolerance below F produces at most a threshold, whatever the gain.

3.5 Bayesian-weight gain: the condition $\kappa > 2$ and the peak

For $G = c\,\Pi/(\Pi + \Pi_0)$, the ratio $G/\Pi = c/(\Pi + \Pi_0)$ decreases in Π, so its supremum is the limit c/Π_0 as $\Pi \to 0$. Condition (5) therefore becomes

$$\Pi_0 < \frac{c\,\lambda}{2\Sigma}. \tag{6}$$

Over-weighting ratio. At low precision the Bayesian-weight gain grows like $(c/\Pi_0)\Pi$, while the optimal gain grows like $F\Pi$. Define

$$\kappa = \frac{c/\Pi_0}{F}.$$

Condition (6) is then $\kappa > 2$. A relay whose prior equals the coordinate's own stationary precision, $\Pi_0 = \lambda/\Sigma$, has $\kappa = c$, so for weak coupling ($c < 2$) a correctly set prior shows no peak.

The criterion $\kappa > 2$ is specific to this family: it holds because G/Π is largest as $\Pi \to 0$. In general the criterion is (5), not the initial slope. Section 3.8 gives a gain with zero initial slope that still opens a window.

Proposition 4. Under (6), σ^2 has exactly one interior maximum. It lies above F and occurs at

$$\Pi^\star = \frac{-B + \sqrt{B^2 - 4AC}}{2A}, \qquad A = 2\Pi_0\Sigma + c^2 + c\lambda, \quad B = 4\Pi_0^2\Sigma, \quad C = \Pi_0^2(2\Pi_0\Sigma - c\lambda).$$

When (6) fails, σ^2 decreases monotonically from F to H. The slope at $\Pi = 0$ is $c(c\lambda - 2\Pi_0\Sigma)/(2\Pi_0^2\lambda^2)$, positive exactly under (6); this is the small-Π form of the excess identity (Appendix A).

Proof. The numerator of $d\sigma^2/d\Pi$ is $-c\,(A\Pi^2 + B\Pi + C)$ (Appendix A). Since $A, B > 0$, there is a unique positive root if and only if $C < 0$, which is (6). □

Corollary (four regimes). Let $M = \sigma^2(\Pi^\star)$. Under (6):

Tolerance R^2	Failure set
$R^2 \le H$	every precision
$H < R^2 \le F$	below a single threshold
$F < R^2 < M$	**a bounded window**
$R^2 \ge M$	none

Without (6), σ^2 decreases strictly from F to H, so only three regimes occur: every precision ($R^2 \le H$), below a single threshold ($H < R^2 < F$), and none ($R^2 \ge F$). The boundary cases sit where shown because, under (6), σ^2 exceeds H at every precision and exceeds F for all small Π.

In the window regime, the variance falls back below the tolerance at high precision because the correction dominates the intrinsic dynamics: the error approaches $H = \Sigma/(\lambda + G_\infty)$, a level set by the gain's saturation rather than by any recovery of the relay.

3.6 Coloured sensory noise

Replace the white sensory noise by an Ornstein–Uhlenbeck process η with correlation time τ, normalized to the same zero-frequency power ($\int \langle \eta(0)\eta(t)\rangle\, dt = 1/\Pi$):

$$d\eta = -\frac{\eta}{\tau}\, dt + \frac{1}{\tau\sqrt{\Pi}}\, dW_s.$$

Why this normalization. White noise has no finite instantaneous variance, so in the white-noise model Π is already the inverse of the zero-frequency noise power. Matching zero-frequency power therefore keeps the meaning of Π unchanged, makes the white-noise model the $\tau \to 0$ limit, and keeps Π the precision of the low-frequency information the relay integrates. Matching variance instead would make the sensory noise vanish as $\tau \to 0$ and would change what Π measures as τ varies.

Proposition 5. With $a = \lambda + G$,

$$\sigma_\tau^2 = \frac{\Sigma + G^2/\big(2\Pi(1+a\tau)\big)}{a}, \tag{7}$$

$$\sigma_\tau^2 - F = \frac{G\big(\lambda G/(2\Pi(1+a\tau)) - \Sigma\big)}{\lambda a}, \qquad \sigma_\tau^2 > F \iff \frac{G}{\Pi} > 2F(1+a\tau). \tag{8}$$

Proof. Solve the stationary Lyapunov equation for (x, η) (Appendix A). □

At matched zero-frequency power, coloured noise injects *less* variance than white noise, because its spectrum lies below the white spectrum at every nonzero frequency. More gain is therefore needed before injected noise outweighs damping. Because $a = \lambda + G$, the threshold in (8) depends on the gain.

The theorem of Section 3.4 carries over, since (7) has the same limits F and H, with (5) replaced by

$$\sup_{\Pi > 0} \left[\frac{G}{\Pi} - 2F(1+a\tau)\right] > 0.$$

For the Bayesian-weight gain, $\frac{d}{d\Pi}\left[G/\Pi - 2F(1+a\tau)\right] = -c(\lambda + 2\Pi_0\Sigma\tau)/\left(\lambda(\Pi+\Pi_0)^2\right) < 0$. The supremum is therefore the $\Pi \to 0$ limit, and for this gain family the window condition is exactly

$$\frac{c}{\Pi_0} > 2F(1+\lambda\tau). \tag{9}$$

Condition (9) uses the monotonicity of the Bayesian-weight gain. For other gains, the condition is the supremum above.

3.7 Discrete time

Consider the sampled plant $h_{k+1} = \phi h_k + w_k$ with $\operatorname{var} w = q$ and $0 < \phi < 1$, observed as $y_k = h_k + v_k$ with $\operatorname{var} v = r = 1/\Pi_d$, where Π_d is the per-sample precision. It is estimated by a constant-gain filter with gain $L \in [0,1]$. Let M_L and S_L be the stationary prior (predicted) and posterior error variances, and let $F = q/(1-\phi^2)$ be the floor at $L = 0$.

Proposition 6.

$$M_L = \frac{\phi^2 L^2 r + q}{1 - \phi^2(1-L)^2}, \qquad S_L = (1-L)^2 M_L + L^2 r, \qquad S_L - F = \frac{M_L - F}{\phi^2},$$

and

$$M_L > F \iff S_L > F \iff \frac{L}{2-L} > F\,\Pi_d. \tag{10}$$

The steady-state Kalman gain $L^\star$ satisfies $L^\star/(2-L^\star) < \frac{1}{2}F\Pi_d$ for every $\Pi_d > 0$.

Proof. See Appendix A. □

For small gains, (10) reads $L > 2F\Pi_d$, the discrete counterpart of (3). The optimal filter again stays more than a factor of 2 below the threshold. In the continuum limit, with $\phi = e^{-\lambda\Delta t}$, $q = \Sigma(1-\phi^2)/\lambda$, $\Pi_d = \Pi\,\Delta t$ and $L \approx G\,\Delta t$, condition (10) reduces to $G > 2F\Pi$. Under the analogous limit hypotheses ($L \to 0$ and $L^2/\Pi_d \to 0$ as $\Pi_d \to 0$, and L tending to a positive limit as $\Pi_d \to \infty$), the theorem of Section 3.4 carries over with $\sup_{\Pi_d}\left[L/(2-L)\right]/\Pi_d > F$ in place of (5).

Proposition 6 is not merely a translation of (3), for three reasons:

- *It is exact at any sampling interval*, not only in the limit $\Delta t \to 0$.
- *It covers both predicted and updated error.* A sampled filter must report one or the other, and the two cross the floor at the same gain.
- *Its threshold is lower than the continuous one suggests.* Because $L/(2-L) > L/2$ for $L > 0$, the exact threshold on the gain, $2F\Pi_d/(1+F\Pi_d)$, lies below the small-gain value $2F\Pi_d$. A sampled relay therefore becomes worse than no filtering at a smaller gain than the continuous result would indicate.

The reason is visible in the recursion. At the floor, one correction with gain L removes a fraction $L(2-L)$ of the error variance, rather than the $2L$ its linearization suggests, and injects measurement noise $L^2 r$. Condition (10) states exactly that injection exceeds removal:

$$L^2 r > F\,L(2-L).$$

A finite correction step removes less variance than its linear approximation while injecting the same noise, so the balance tips at a smaller gain. The continuous counterpart is (3), which compares injection $G^2/(2\Pi)$ with removal $\Sigma G/\lambda$.

3.8 Other gain shapes

A gain that ignores precision. Let $G = G_0$ be constant. Then $d\sigma^2/d\Pi = -G_0^2/\big(2\Pi^2(\lambda + G_0)\big) < 0$, and $\sigma^2 \to \infty$ as $\Pi \to 0$. There is no window: failure is a lower threshold, with unbounded error below it.

A gain with no initial over-weighting. For the Hill gain $G = c\,\Pi^2/(\Pi^2 + \Pi_0^2)$, the initial slope is zero. However, G/Π peaks at $\Pi = \Pi_0$ with value $c/(2\Pi_0)$, so (5) holds whenever $\Pi_0 < c\lambda/(4\Sigma)$, and a window then opens at intermediate precision. Initial over-weighting is sufficient for a window but not necessary.

A capped precision schedule. A gain that rises linearly with precision and then saturates, $G = \min(g\Pi, c)$, has $G/\Pi = g$ up to $\Pi = c/g$ and smaller beyond. It therefore opens a window exactly when $g > 2F$. A filter whose gain does not depend on precision is the precision-blind case above and cannot open one.

4. Worked Example

Parameters: $\lambda = c = 1$, $\Sigma = 0.01$. This gives $F = 0.01$ and $H = 0.005$, and condition (6) reads $\Pi_0 < 50$.

How much miscalibration this example contains. The miscalibrated relay below has $\Pi_0 = 1$, so $\kappa = 100$: its gain initially rises a hundred times faster than the optimal gain. This is an extreme case, chosen to make the window visible. Smaller over-weighting gives much smaller peaks:

Over-weighting κ	2	3	5	10	20	50	100
Peak / F	1.000	1.036	1.175	1.584	2.432	5.000	9.288

At $\kappa = 50$ ($\Pi_0 = 2$), the peak is exactly $5F = 0.05$, at $\Pi^\star = 4/3$. In this example, then, $R^2 = 0.05$ is failed only when $\kappa > 50$ (Figure 3). A threefold over-weighting raises the variance by at most 3.6%.

Miscalibrated relay, $\Pi_0 = 1$ ($\kappa = 100$). The peak is $M = 0.09288$ at $\Pi^\star = 0.6867$.

- For $R^2 = 0.05$, the failure set is exactly $\Pi \in (1/9,\ 4)$.
- For $R^2 = 0.008$, the failure set is $\Pi < 82.67$, a threshold.

Calibrated relay, $\Pi_0 = \lambda/\Sigma = 100$ ($\kappa = 1$). Variance decreases monotonically: 0.01000, 0.009950 and 0.007500 at $\Pi = 0.01$, 1 and 100.

Optimal filter, Eq. (4). Variance is 0.01000, 0.009950 and 0.007321 at the same precisions, decreasing to 0.

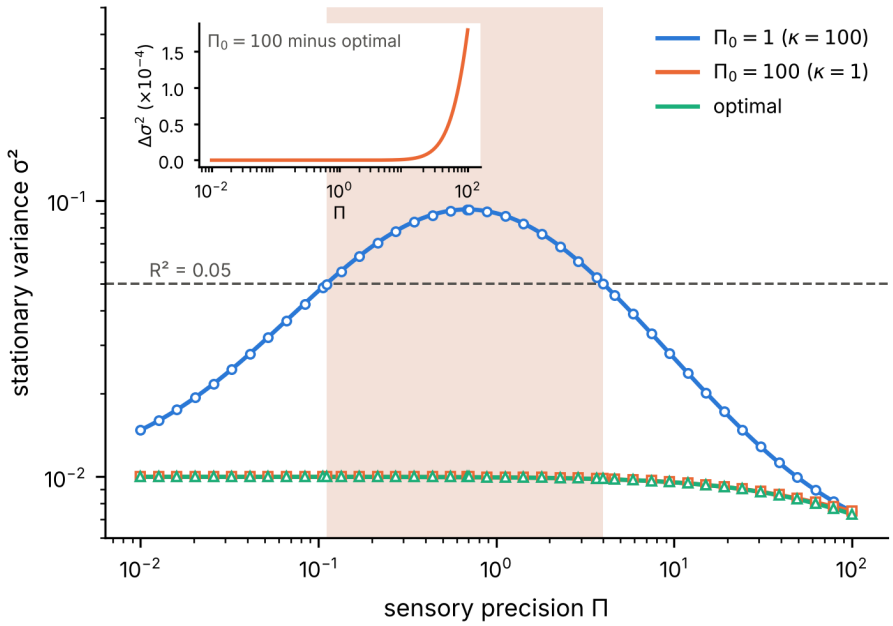


Figure 1. Stationary variance of the miscalibrated ($\Pi_0 = 1$, $\kappa = 100$), calibrated ($\Pi_0 = 100$, $\kappa = 1$) and optimal relays. Lines are closed forms; points are Euler–Maruyama estimates (± 1 SE) on identical noise. The dashed line marks $R^2 = 0.05$, and the shaded band is the window $(1/9, 4)$, which spans a factor of 36 in precision. The inset shows calibrated minus optimal variance, which stays below 10^{-5} up to $\Pi \approx 20$ and about 1.8×10^{-4} at $\Pi = 100$.

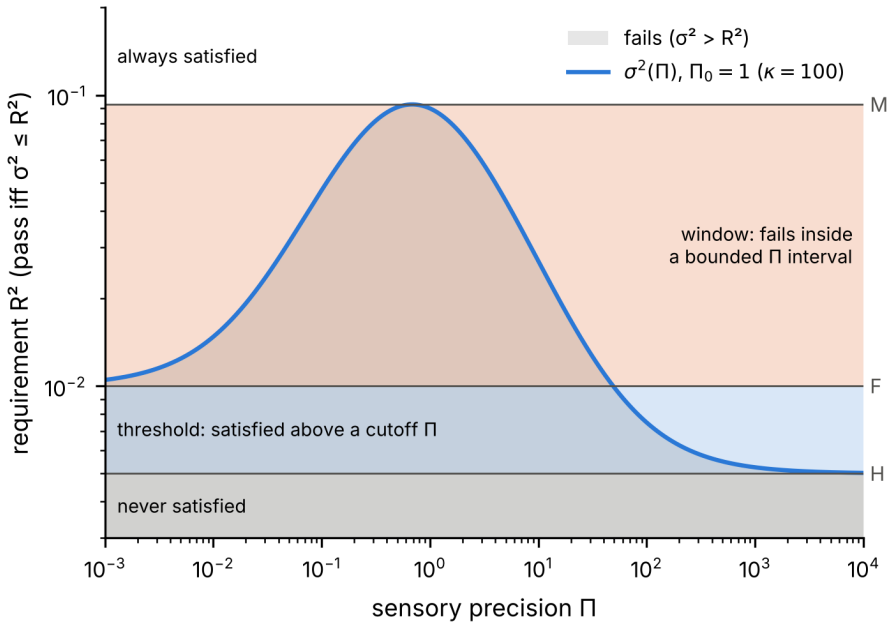


Figure 2. Failure regime as a function of R^2 for $\Pi_0 = 1$. The boundaries are at H, F and M.

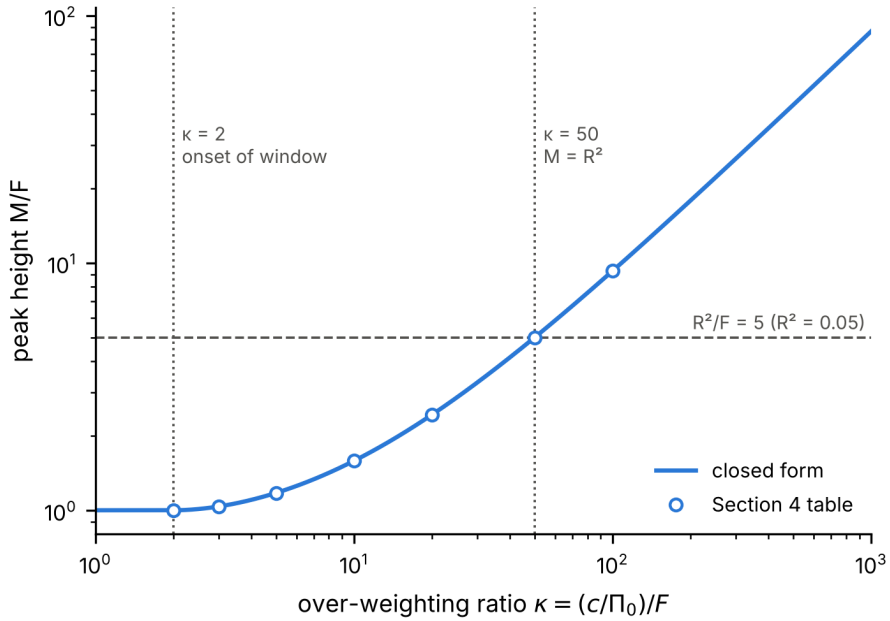


Figure 3. Peak height M/F of the Bayesian-weight relay against the over-weighting ratio κ (closed form). A window first appears at $\kappa = 2$. The worked example's tolerance, $R^2/F = 5$, is reached at $\kappa = 50$. Points are the values in the table above.

5. Verification

The analytical results do not rest on simulation. The symbolic and numerical checks below were specified in advance as verification checks, with pass/fail criteria committed before the corresponding runs and not changed afterward. Every check passed except one grid-resolution check, which is reported, with commits and seeds, in Appendix B. All code and results are in the accompanying repository [8].

5.1 Confirmatory checks

Symbolic. SymPy confirms the following:

- Eqs. (2)–(4), including Proposition 3 from the Riccati equation;
- the floor and high-precision limits;
- the derivative numerator and root condition of Proposition 4, and the small-Π slope;
- Proposition 5;
- every part of Proposition 6: M_L derived from the recursion, $S_L - F = (M_L - F)/\phi^2$, the factorization, the equivalence (10) (with no mismatch at 10,000 exact rational parameter sets), the optimal-gain inequality, and the continuum limit, whose first-order term in Δt is $G/2 - F\Pi$;
- the Section 4 values, including $\Pi^\star = 4/3$ and $M = 1/20$ exactly at $\kappa = 50$, and $\kappa = 50$ as the smallest over-weighting that reaches $M = 0.05$.

Simulation. Euler–Maruyama simulation ($dt = 10^{-3}$, 256 trajectories, 200 time units after burn-in) matches Eq. (2) within 0.29% at 43 precisions. These checks confirm the closed forms. Because the model is an Ornstein–Uhlenbeck process, they carry little evidential weight beyond the algebra.

Paired comparison. On identical noise, the miscalibrated relay crosses $R^2 = 0.05$ at 0.1118 and 3.989 (exact values $1/9$ and 4). For the calibrated and optimal relays, the pre-registered test was one-sided: failure if any point exceeded the same relay's value at the lowest precision, $\Pi = 0.01$, by more than 3 standard errors of the paired difference. No point exceeded it. The largest standardized difference (point minus lowest-precision value) was -1.95.

5.2 Exploratory checks: breaking one assumption at a time

These checks were exploratory. Predictions were stated in advance, and outcomes are reported either way. Two outcomes are reported separately:

- a **peak above** F, equivalent by the theorem to a window existing for some tolerance;
- the $R^2 = 0.05$ **interval**, the failure set at the worked example's tolerance.

Discrete time. The setup:

- The true plant samples the continuous worked example at $\Delta t = 0.05$, with $\phi = e^{-\lambda \Delta t}$ and $q = \Sigma(1 - \phi^2)/\lambda$, so its floor is $F = 0.01$.
- The swept variable is the per-sample precision $\Pi_d = 1/r$, which corresponds to $\Pi = \Pi_d/\Delta t$.
- Each filter uses the steady-state Kalman gain of an *assumed* model whose noise ratio is wrong by a factor $k \in \{1, 2, 3, 10, 100\}$.
- The "slope ratio" is the gain's initial slope $dL/d\Pi_d$ divided by F, the slope of the correct filter. It equals k, and because $L = G\Delta t$ and $\Pi_d = \Pi\Delta t$, it is exactly the continuous over-weighting ratio κ.
- The reported variance is the posterior error variance S_L.

These filters belong to a different gain family from Section 4, so the table below should be compared with Section 4 for onset, not for peak height. The pre-stated prediction was that a peak above F appears if and only if the slope ratio exceeds 2. It was correct in all 10 cases, including the boundary at ratio 2. Proposition 6 accounts for the onset: near $\Pi_d = 0$, condition (10) reduces to slope ratio > 2, and it applies equally to prior and posterior variance.

Discrete filters with the Kalman gain of a misspecified model (a different gain family from Section 4); the interval is in per-sample precision Π_d:

Slope ratio	1	2	3	10	100
Peak above F	no	no	yes	yes	yes
Peak height	—	—	0.01075	0.02224	0.1837
$R^2 = 0.05$ interval	none	none	none	none	$(0.0046, 7.29)$

Qualitatively as in Section 4, modest miscalibration gives a small peak, and only the factor-100 case fails $R^2 = 0.05$. Closed-form values from Proposition 6 reproduce all three reported peaks to within 2×10^{-6} relative.

The heights in this table are *not* comparable with Section 4 at the same κ. These filters share the paper's initial slope but not its gain family. A Kalman gain computed from an inflated noise ratio keeps rising with precision instead of saturating at c, so it admits more sensory noise near the peak. The ratio-100 peak, 0.1837, against 0.0929 for the Bayesian-weight relay at $\kappa = 100$, decomposes almost entirely into:

- gain shape: $+0.083$, since the continuous-time limit of these filters peaks at 0.1758;
- finite Δt: $+0.008$.

Prior variance (peak 0.1672) and posterior variance differ only at finite Δt, and the gap shrinks to 3×10^{-4} at $\Delta t = 0.001$. Only the onset condition carries over between gain families; peak height depends on the gain's shape.

Inflating the assumed process noise and inflating the assumed measurement precision give identical filters at equal factors, so the two arms are one test.

Coloured sensory noise. Simulations match Eq. (7) (max $|z| \leq 0.97$). Values from (7) at $\Pi_0 = 1$:

τ	0.01	0.1	1	10
Peak	0.09169	0.08234	0.04326	0.01356
Peak location	0.683	0.655	0.538	0.330
$R^2 = 0.05$ interval	$(0.113, 3.90)$	$(0.130, 3.14)$	none	none

A peak above F survives at every tested τ, but the $R^2 = 0.05$ interval disappears once τ approaches the relay time constant.

The pre-registered symbolic check (Task 6) confirmed (7), (8) and the slope condition. At $\Pi_0 = 1$ that condition predicts a rise iff $\tau < 49$; this held at $\tau = 48, 48.9, 49.1$ and 50, and no excess appeared at $\tau = 50, 100, 1000$.

Drifting signal. The signal mean drifted either sinusoidally (amplitudes 0.05 and 0.1, periods 10 and 100) or as a slow random walk. In every case the $R^2 = 0.05$ interval persists with its lower edge unchanged; its upper edge moves out to at most 4.15 under sinusoidal drift and 4.38 under the random walk. Drift adds a second failure mode at high precision: because the relay passes drift with gain $G/(\lambda + G)$, the high-precision variance rises. On the grid used for this check, the $R^2 = 0.008$ threshold moves from 83.0 without drift (exact value 82.67) to 133–142 for the larger sinusoid. Under random-walk drift, variance about z^* grows with run length and has no stationary value, so no high-precision number is reported.

Other gains. Gains $c \tanh(\Pi/\Pi_0)$ and $c(1 - e^{-\Pi/\Pi_0})$ were swept over 61 values of Π_0. For both, a rise near $\Pi = 0$ occurred exactly when $\Pi_0 < 50$, and a peak exactly when a rise did. The Hill gain never rose near $\Pi = 0$, but peaked above F for every tested Π_0 up to 19.95. Its exact boundary is $\Pi_0 = 25$. Checks on either side confirm it: the maximum excess is 1.40×10^{-4} at $\Pi_0 = 24$, exactly 0 at $\Pi_0 = 25$, and no precision exceeds F at $\Pi_0 = 26$. Every failure set was a single interval.

Precision-blind gain. For constant $G_0 \in \{0.1, 0.5, 1, 2\}$, variance was monotone decreasing and diverged as $\Pi \to 0$.

6. Relation to Prior Work

The related work divides cleanly. Formulas for the actual error of a filter with a wrong gain or noise model are old, and Eq. (2) is the scalar steady state of them. What we did not find is the question this note asks of those formulas: how the set $\{\Pi : \sigma^2(\Pi) > R^2\}$ is shaped as observation precision is swept, and when it is a bounded window. The neighbours below either supply the

formulas or report non-monotone error against some other swept variable.

- **Mismatched filtering, discrete and continuous.** Heffes [7] and Friedland [3] give the excess error of filters with wrong noise statistics or gains. Nishimura [10] and Griffin and Sage [6] derive the actual-error covariance equations for continuous-time suboptimal filters, together with bounds. Nishimura shows that the computed covariance bounds the actual one when the assumed noise covariances are at least the true ones, and notes that optimistic choices are especially harmful. Over-weighting the observation corresponds to such an optimistic measurement model, outside that conservative bound. Sangsuk-Iam and Bullock [12], Teunissen et al. [14] and Crouse [2] extend the actual-error analysis in discrete time, and Ge, Shao, Duan and Wen [5] rank the ideal, filter-calculated and true mean-squared errors under mismatched noise covariances. Eq. (2) is the scalar steady state of these equations, and Proposition 3 is the classical Kalman–Bucy result [9].
- **Constant-gain filtering.** That an over-gained constant-gain filter can do worse than no filtering is the familiar noise-versus-responsiveness trade-off. The present note supplies its exact boundary, $G = 2F\Pi$, for this plant.
- **Optimal feedback strength in gene networks** [13]. Noise can be minimized at an intermediate feedback strength. The swept variable there is feedback strength, not observation precision, and there is no tolerance analysis.
- **When feedback is useful** [1]. The optimal gain of predictive feedback depends non-monotonically on computational parameters. This work asks when feedback is worth using, not how a given relay fails as precision varies.
- **Non-monotone variance under misspecification** [11]. In a Kalman smoother with a misspecified model-error decorrelation time, the *filter-reported* posterior variance can be concave in the guessed timescale, while the actual mean-square error is convex. The swept variable and the non-monotone quantity both differ from those here.
- **Precision weighting in predictive coding** [4] motivates Bayesian-weight gains, but it does not give a tolerance analysis.

Contribution. Within a scalar, stationary, linear filtering model, the note characterizes the topology of the tolerance-failure set as observation precision varies; it claims no new excess-error formulas. To our knowledge, prior analyses have not identified failure confined to a bounded interval of observation precision under a fixed tolerance, condition (5) for its existence, or the fact that the optimal filter cannot exhibit it. The supporting results are:

- the excess identity (3) and its discrete analogue (10);
- the Bayesian-weight specialization (6), with its unique peak;
- the dependence of peak height on over-weighting (Section 4);
- the coloured-noise extension (Proposition 5).

This assessment rests on targeted searches of the estimation, control and data-assimilation literature, not an exhaustive one.

7. Scope

The analytical results assume:

- linear scalar dynamics;

- Gaussian noise, with white process noise and sensory noise that is white or Ornstein–Uhlenbeck (Proposition 5);
- a gain that is constant in time at each precision;
- a signal centred on the target;
- a user-chosen tolerance.

Discrete time is covered by Proposition 6.

The diagnostic reading holds only within this class. For linear, scalar, stationary relays of this form, failure confined to intermediate precision means the gain exceeded $2F\Pi$ somewhere (under white sensory noise) or the gain-dependent threshold $2F(1+a\tau)\Pi$ of Eq. (8) (under coloured noise). Outside the class this need not hold. Drift already produces a separate high-precision failure mode (Section 5.2), and nonlinear dynamics, time-varying gains or multivariate states could plausibly produce windows by other routes.

Multivariate states. For a vector state with a matrix gain, the analogue of (2) is a Lyapunov equation, and the excess over the floor (the open-loop covariance) can be positive in some directions and negative in others. Whether (3) has a clean matrix form, such as a condition on the gain relative to $2F\Pi$ in each eigendirection, is left open.

Nothing here claims that any physical or biological system implements this relay.

Appendix A. Derivations

Derivative of (2), Bayesian-weight gain. Write $\sigma^2 = N/D$ with

$$N = \Sigma(\Pi+\Pi_0)^2 + \tfrac{1}{2}c^2\Pi, \qquad D = (\Pi+\Pi_0)\big((\lambda+c)\Pi+\lambda\Pi_0\big).$$

The derivative $d\sigma^2/d\Pi$ has denominator $2(\Pi_0+\Pi)^2(\Pi_0\lambda+\Pi c+\Pi\lambda)^2$ and numerator

$$-2\Pi_0^3\Sigma c - 4\Pi_0^2\Pi\Sigma c + \Pi_0^2c^2\lambda - 2\Pi_0\Pi^2\Sigma c - \Pi^2c^3 - \Pi^2c^2\lambda = -c\big(A\Pi^2+B\Pi+C\big).$$

Excess identity.

$$\sigma^2 - \frac{\Sigma}{\lambda} = \frac{\lambda\Sigma+\lambda G^2/(2\Pi)-\Sigma(\lambda+G)}{\lambda(\lambda+G)} = \frac{G\big(\lambda G/(2\Pi)-\Sigma\big)}{\lambda(\lambda+G)}.$$

Small-Π expansion. With $G = g_0\Pi + O(\Pi^2)$,

$$\sigma^2 = F + \Pi\left(\frac{g_0^2}{2\lambda} - \frac{\Sigma g_0}{\lambda^2}\right) + O(\Pi^2).$$

This is positive iff $g_0 > 2F$. For the Bayesian-weight gain $g_0 = c/\Pi_0$, so the condition is $\kappa > 2$.

Optimal gain. Minimizing (2) over G gives $G^2 + 2\lambda G - 2\Sigma\Pi = 0$, the same root as the Riccati equation with $G = p\Pi$. At the root, $\sigma^2 = G/\Pi = p$.

Coloured noise. For the state (x,η), the drift matrix is $\begin{pmatrix} -a & G \\ 0 & -1/\tau \end{pmatrix}$ and the noise covariance is $\mathrm{diag}\big(2\Sigma,\ 1/(\tau^2\Pi)\big)$. The stationary Lyapunov equation gives

$$C_{\eta\eta} = \frac{1}{2\Pi\tau}, \qquad C_{x\eta} = \frac{G}{2\Pi(1+a\tau)}, \qquad C_{xx} = \frac{\Sigma + G\,C_{x\eta}}{a},$$

which is (7). With the Bayesian-weight gain,

$$\sigma_\tau^2 = F + \frac{c\big(c\lambda - 2\Pi_0\Sigma(1+\lambda\tau)\big)}{2\Pi_0^2\lambda^2(1+\lambda\tau)}\,\Pi + O(\Pi^2).$$

Discrete time (Proposition 6). The predicted error obeys

$$e_{k+1}^- = \phi\big[(1-L)\,e_k^- + Lv_k\big] - w_k,$$

so the stationary variance satisfies $M_L = \phi^2\big[(1-L)^2 M_L + L^2 r\big] + q$, which gives M_L. Direct factorization gives

$$M_L - F = \frac{\phi^2 L\,\big[L\big(r(1-\phi^2)+q\big) - 2q\big]}{(1-\phi^2)\big(1+\phi(1-L)\big)\big(1-\phi(1-L)\big)}.$$

The denominator is positive for $L \in [0,1]$, so $M_L > F$ iff $L > 2q/\big(r(1-\phi^2)+q\big) = 2F\Pi_d/(1+F\Pi_d)$, which is equivalent to $L/(2-L) > F\Pi_d$. The same factorization holds for $S_L - F$ with ϕ^2 removed from the numerator, so $S_L - F = (M_L - F)/\phi^2$.

For the optimal gain, write $s = 1-\phi^2 \in (0,1)$. The steady-state Riccati equation is $M = \phi^2 Mr/(M+r) + q$, so $q = M(M+rs)/(M+r)$, and $L^\star = M/(M+r)$ gives $L^\star/(2-L^\star) = M/(M+2r)$. The claim $M/(M+2r) < \frac{1}{2}F\Pi_d = q/(2rs)$ is then equivalent to

$$2rs(M+r) < (M+rs)(M+2r),$$

whose difference is $M^2 + rM(2-s) > 0$. SymPy confirms the identity. Across 20,000 random parameter sets, the largest value of $\big[L^\star/(2-L^\star)\big]\,/\,\big(\frac{1}{2}F\Pi_d\big)$ was 0.9999993, approaching the bound only as $\Pi_d \to 0$.

Hill gain. $G/\Pi = c\,\Pi/(\Pi^2 + \Pi_0^2)$ is maximized at $\Pi = \Pi_0$ with value $c/(2\Pi_0)$, so (5) becomes $\Pi_0 < c\lambda/(4\Sigma)$, which is 25 in the worked example.

Peak at $\kappa = 50$. With $\Pi_0 = 2$: $A = 2.04$, $B = 0.16$ and $C = -3.84$, so $\Pi^\star = 4/3$. Then $K_s = 0.4$, and $\sigma^2 = (0.01 + 0.06)/1.4 = 0.05$ exactly.

Appendix B. Reproducibility

The verification suite is the `verification/` directory of the accompanying repository [8], `github.com/RedWing-Duck/Bounded-Failure-Window`, which will be made public on publication. It contains:

- `PREREG.md`, `PREREG_t6.md` and `PREREG_t7.md`: the pre-registered criteria, committed before the corresponding runs and not edited since;
- scripts `t1_symbolic.py` to `t4e_fixed_gain.py`, `t6_colored_symbolic.py` and the `t7_*.py` scripts, one per task;

- `results/`: JSON outputs, logs and text reports, including `env.json` (environment) and `t7_4a_spec.txt` (the Task 4a specification used in Section 5.2);
- `figures/`: all figures;
- `REPORT.md`: every criterion with its measured values, generated from `results/`;
- `run_all.sh`: regenerates everything from scratch.

`run_all.sh` regenerates every number and figure in about 11 minutes (679 s wall time on 4 CPUs). A rerun in a fresh copy reproduced all 13 result files and all 11 figures exactly.

- **Environment:** Python 3.13.16, NumPy 2.5.3, SciPy 1.18.1, SymPy 1.14.0, Matplotlib 3.11.2; Linux, 4 CPUs.
- **Random numbers:** NumPy's PCG64 generator (`default_rng`), float64. Within each script, common random numbers drive every precision and every relay, so simulated comparisons are paired.
- **Seeds:** Task 2: 1002; Task 3: 1003; Task 4a: 1041; Task 4b: 1042; Task 4c: 1043; Task 7: 7001 and 7002 (random parameter sets for Proposition 6), with 7003 for the symbolic fallback. Tasks 1, 4d, 4e and 6 are deterministic.
- **Simulation settings:** Euler–Maruyama with $dt = 10^{-3}$; 256 trajectories per precision; 20 time units of burn-in and 200 of sampling; batch-means standard errors with batches of 10 time units (10 relaxation times); 43 precisions (40 log-spaced points in $[10^{-2}, 10^{2}]$ plus $1/9$, 0.687 and 4).
- **Commits:** pre-registration `d21f193`; results `70ebb55`; Task 6 pre-registration `824ca06`; Task 7 pre-registration `a0d2be2`, results `2f9da71`.
- **Figures in this paper:** Figures 1, 2 and 3 are, in `figures/`, `fig1_paired.png`, `fig2_regimes.png` and `fig3_peak_vs_kappa.png`, all written by `t7_figures.py`. Task 3 also writes `fig1a_paired.png` and `fig1b_regimes.png`, the same content in the earlier notation (P for Π_0); they remain part of the original Task 3 output.

The one failed criterion. A Task 6 criterion required peak locations computed from (7) to match Task 4b's to three significant figures. All peak heights matched, but three locations differed in the third figure (for example, 0.683 vs 0.684). Task 4b had located peaks on a grid with steps of about 0.4%; on that same grid, (7) reproduces its values to 10^{-12}. The discrepancy is grid resolution, not the formula. The criterion is recorded as failed and was not changed.

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